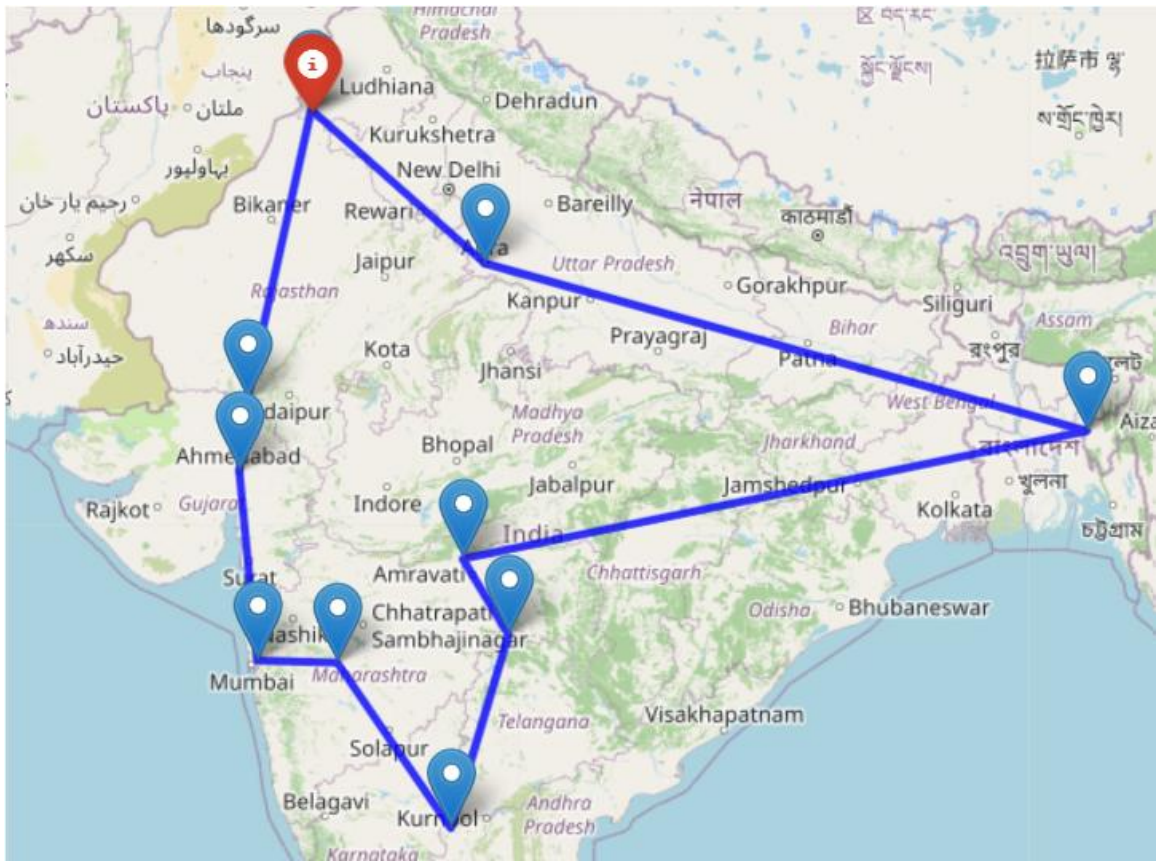




Understanding the Traveling Salesman Problem (TSP)

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Introduction

Have you ever wondered how online deliveries are planned?

Every day, thousands of orders are delivered to customers through services like Amazon, Flipkart, Swiggy and Blinkit. Yet a delivery partner doesn't carry just one package. Instead, they leave a nearby hub with multiple orders, visit several customers along an efficient route, and return to the hub after completing their deliveries or carrying back packages that couldn't be delivered.

This seemingly simple task is powered by one of the most well-known optimization problems in Operations Research: the Traveling Salesman Problem (TSP).

The objective of TSP is straightforward: determine the shortest possible route that visits every required location exactly once and returns to the starting point. While the idea sounds simple, finding the optimal route becomes surprisingly difficult as the number of locations increases.

WHY IS TSP CHALLENGING?

TSP belongs to the class of NP-Hard problems. This doesn't mean it's impossible to solve but as the number of cities increase, the number of possible routes grow extremely rapidly. Verifying the length of a given route is straightforward but finding the absolute best route requires searching through an enormous number of possibilities.

This is why optimization techniques are essential for solving real-world routing problems efficiently.

SOLUTION TECHNIQUES

To understand how different optimization approaches influence solution quality and computation time, I experimented with several techniques during my implementation. Each method approaches the same problem differently, offering unique trade-offs between computational speed and solution quality.

- **GUROBI: SEARCHING FOR THE BEST POSSIBLE ROUTE**

It is a mathematical optimization solver that explores possible routes to find the best solution. Instead of randomly checking all the routes, it systematically eliminates routes that can't lead to better routes, allowing it to search in a better way.

If the solver reaches the specified time limit before proving that the solution is globally optimal, it still returns the **best route found so far** along with the **MIP Gap**, which indicates how close the current solution is to the theoretical optimum.

- **OR-TOOLS: A FAST FEASIBLE SOLUTION**

It focuses on finding a high-quality route as quickly as possible rather than proving it is globally optimal.

It efficiently generates a route that works well within a short amount of time, making it an excellent choice when obtaining a good solution quickly is more important than guaranteeing the absolute best one.

- **WARM START: BEGINNING WITH A GOOD SOLUTION**

Optimization algorithms often spend considerable time searching for an initial feasible route.

Warm Start improves this process by supplying the solver with a good initial solution before the optimization begins.

In my implementation, Warm Start allowed the optimization process to focus more on improving the existing solution rather than spending time searching for its first feasible route.

- **KNN SPARSIFICATION: REDUCING THE SEARCH SPACE**

One of the biggest challenges in solving large TSP instances is the enormous number of possible connections between cities.

KNN (K-Nearest Neighbour) Sparsification reduces this complexity by allowing each city to connect only with its nearest neighbouring cities instead of every city in the network.

The value of K determines how large this neighbourhood is. A smaller K results in faster computation but may miss better routes, while a larger K explores more possibilities at the cost of increased computation time.

KEY TAKEAWAYS

- The Traveling Salesman Problem models many real-world routing and logistics applications.
- TSP is considered NP-Hard because the number of possible routes grows extremely rapidly as the problem size increases.
- Gurobi focuses on finding the best possible solution and provides information about how close the solution is to optimality.
- OR-Tools prioritizes quickly generating a high-quality feasible solution.
- Warm Start improves optimization by providing a strong initial solution instead of starting from scratch.
- KNN Sparsification reduces computational effort by limiting the search to nearby locations while maintaining solution quality.